Submitted By:

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Assignment:

Project No. 1

Submitted To:
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For the fulfillment of,
Object-Oriented Programming Lab
Department of Computer Engineering
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TASK:

Create a Pacman game using OOP that involves the use of multiple classes and objects that interact with each other. The PackmanGame class contains the main method and initializes the game by creating instances of the GameBoard, Pacman, Ghost, and Food classes. The PacmanGame class also contains the game loop and handles user input, allowing the player to control Pacman's movement around the board.

SOLUTION:

```
package Project1;
import java.util.Scanner;
import java.util.Random;
import java.util.*;
class GameBoard {
  private int size = 10;
  private char[][] maze = {
      {'#','#','#','#','#','#','#','#','#','#'},
      {'#','.','.','.','.','.','.','.','.','#'},
      {'#','.','#','#','.','#','#','.','.','#'},
      {'#','.','.','#','.','.','#','.','.','#'},
      {'#','#','.','#','#','.','#','#','.','#'},
      {'#','.','.','.','.','.','.','#','.','#'},
      {'#','.','#','#','#','#','.','#','.','#'},
      {'#','.','.','.','.','#','.','.','.','#'},
      {'#','.','.','#','.','.','.','#','.','#'},
      {'#','#','#','#','#','#','#','#','#','#'}
  };
  public void display(Pacman p, Ghost g) {
    for (int i=0;i<size;i++) {
      for (int j=0;j<size;j++) {
        if (p.getX()==i && p.getY()==j) {
          System.out.print("P ");
        }
        else if (g.getX()==i && g.getY()==j) {
          System.out.print("G ");
        }
        else {
          System.out.print(maze[i][j] + " ");
        }       }
      System.out.println();
    }   }
  public boolean isWall(int x, int y) {
    return maze[x][y]=='#';
  }
```

```
    public void checkFood(Pacman p) {
        if (maze[p.getX()][p.getY()]=='.') {
            maze[p.getX()][p.getY()]=' ';
            p.increaseScore();
        }   }
    public int getSize() {
        return size;
    }
}
class Pacman {
    private int x, y;
    private int score;
    public Pacman(int x, int y) {
        this.x=x;
        this.y=y;
        score=0;
    }
    public void move(int dx, int dy) {
        x +=dx;
        y +=dy;
    }
    public void increaseScore() {
        score++;
    }
    public int getScore() {
        return score;
    }
    public int getX() {return x; }
    public int getY() {return y; }
}
class Ghost {
    private int x, y;
    private Random rand=new Random();
    public Ghost(int x, int y) {
        this.x = x;
        this.y = y;
    }
    public void randomMove() {
        int dx=rand.nextInt(3)-1;
        int dy=rand.nextInt(3)-1;
        x += dx;
        y += dy;
    }
    public void setPosition(int x, int y) {
        this.x = x;
```

```java
        this.y = y;
    }
    public int getX() {return x; }
    public int getY() {return y; }
}
class Food {
    private int x, y;
    private boolean eaten;
    public Food(int x,int y) {
        this.x=x;
        this.y=y;
        eaten=false;
    }
    public void eat() {
        eaten=true;
    }
    public boolean isEaten() {
        return eaten;
    }}
public class PacmanGame {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        GameBoard board=new GameBoard();
        Pacman pacman=new Pacman(1, 1);
        Ghost ghost=new Ghost(8, 8);
        boolean running=true;
        while (running) {
          System.out.println("\nScore: " + pacman.getScore());
          board.display(pacman, ghost);
         System.out.print("Move (W/A/S/D): ");
          char move=sc.next().toUpperCase().charAt(0);
          int dx=0,dy=0;
          switch (move) {
              case 'W':dx=-1; break;
              case 'S':dx=1; break;
              case 'A':dy=-1; break;
              case 'D':dy=1; break;
              default: System.out.println("Invalid Input");
          }
          int newX=pacman.getX() + dx;
          int newY=pacman.getY() + dy;
          if (newX >= 0 && newX < board.getSize() &&
                  newY >= 0 && newY < board.getSize() &&
                  !board.isWall(newX, newY)) {
              pacman.move(dx, dy);
```

```
        }
        int oldX=ghost.getX();
        int oldY=ghost.getY();
        ghost.randomMove();
        if (ghost.getX() < 0 ||ghost.getX()>=board.getSize() ||
                ghost.getY()<0 ||ghost.getY()>=board.getSize() ||
                board.isWall(ghost.getX(), ghost.getY())) {
            ghost.setPosition(oldX, oldY);}
        board.checkFood(pacman);
        if (pacman.getX()==ghost.getX() &&
                pacman.getY()==ghost.getY()) {
            System.out.println("Game Over!");
            running=false;
        }      }
    System.out.println("Final Score: " + pacman.getScore());
    sc.close();
}}
```

OUTPUT:

Run PacmanGame

```
# # # # # # # # # #
# .               #
# . # # . # # .   #
# . . # . . # .   #
# # . # # . # #   #
# . . . . . . #   #
# . # # # # . # P #
# . . . . # . . G #
# . . # . . . # . #
# # # # # # # # # #
Move (W/A/S/D): s
Game Over!
Final Score: 13

Process finished with exit code 0
```
